Serum uric acid level as an independent risk factor for all-cause, cardiovascular, and ischemic stroke mortality: a Chinese cohort study.

OBJECTIVE: The association between hyperuricemia and cardiovascular events has been documented in high-risk groups, but is still undetermined in general populations, especially Chinese. This study assessed the temporal association between serum uric acid level, hyperuricemia, and cardiovascular mortality. METHODS: A prospective cohort study of 41,879 men and 48,514 women ages > or = 35 years was conducted using data from the MJ Health Screening Centers in Taiwan. Mortality from all causes, total cardiovascular disease (CVD), ischemic stroke, congestive heart failure, hypertensive disease, and coronary heart disease were compared according to increasing serum uric acid levels. RESULTS: A total of 1,151 (21.2%) events of 5,427 total deaths were ascribed to CVD (mean followup 8.2 years). Hazard ratios (HRs) for hyperuricemia (serum uric acid level >7 mg/dl) were estimated with Cox regression model after adjusting for age, sex, body mass index, cholesterol, triglycerides, diabetes, hypertension, heavy cigarette smoking, and frequent alcohol consumption. In all patients, HRs were 1.16 (P < 0.001) for all-cause mortality, 1.39 (P < 0.001) for total CVD, and 1.35 (P = 0.02) for ischemic stroke. In subgroup analysis, the HRs for cardiovascular risk remained significant in patients with hypertension (1.44, P < 0.001) and in patients with diabetes (1.64, P < 0.001). In addition, in a low metabolic risk subgroup, the HRs for all-cause mortality and total cardiovascular morbidity were 1.24 (P = 0.02) and 1.48 (P = 0.16), respectively. CONCLUSION: Hyperuricemia was an independent risk factor of mortality from all causes, total CVD, and ischemic stroke in the Taiwanese general population, in high-risk groups, and potentially in low-risk groups.